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Thermodynamic analysis of Al–Sc, Cu–Sc and Al–Cu–Sc system

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ABSTRACT

Encouraged by the promising application of the Al–Cu–Sc system in the development of high strength aluminum alloy, the Al-rich region of this system was assessed by the CALPHAD approach. In order to make the models used for the phases in the three binary sub-systems compatible with each other, Al–Sc and Cu–Sc system were remodeled. Substitutional solution model was used for solution phases and sublattice model for binary intermetallic phases and the ternary compound $Al_{8-x}Cu_{4+x}Sc$. Comparisons between the calculated and experimental results showed that most of the experimental information was satisfactorily described by the present thermodynamic description.

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1. Introduction

Scandium is an attractive element added into aluminum alloy to improve strength at elevated temperatures. Knipling et al. [1] summarized four stand-out characteristics of it: (1) A thermodynamically stable strengthening phase Al_3Sc with a $L1_2$ crystal structure is formed in this system. The dispersive nanoparticles of Al_3Sc efficiently prevent the motion of dislocation and grain and sub-grain boundary, then improve the recrystallization temperature and creep resistance; (2) The phase diagram of Al–Sc system exhibits a shallow solvus curve beneficial to precipitation strengthening; (3) The eutectic reaction formed from Liquid to Fcc_Al and Al_3Sc is conducive to casting; (4) The solid–liquid partition coefficient k_0 (C_{max}/C_L, C_{max} and C_L are solid and liquid solubility at the invariant temperature, respectively) approaching unity is a particularly nice trait which leads to a small degree of microsegregation during casting. Usually, Sc is added into the aluminum alloy along with Zr to achieve better mechanical properties with the formation of Al_3(Sc, Zr) phase, and a low-cost as well. As investigated by Dai et al. [2], joint addition of Sc and Zr to the Al–Zn–Mg–Cu based aluminum alloy obviously increased the strength of the alloy with the ductility remaining on a high level. According to their report, the tensile strength was 812.4 MPa and the elongation 6.2% after T6 (120 °C × 24 h) aging. After retrogression and reaging (120 °C × 24 h + 180 °C × 20 min + 120 °C × 24 h), they were 751.2 MPa and 6.4%, respectively. Researches about the properties of ultra-high strength aluminum alloys containing Sc can also be found in other literatures [3–5].

It is worth of noting that the process of accelerated material design depends upon a thorough knowledge of phase diagram of the interested alloy. It will become a helpful guide for selecting alloy composition and optimizing processing parameters. Since Al–Cu–Sc system is an important sub-system of the aluminum alloy containing Sc, the Al-rich corner of this ternary system was assessed by means of the CALPHAD approach. The Al–Sc and Cu–Sc systems have been thermodynamically assessed with the liquid phase using quasi-chemical model and ideal associated solution model, respectively. In this work, both of them were remodeled using substitutional solution model for the sake of model compatibility.

2. Literature review

2.1. The Al–Sc binary system

Naumkin et al. [6] have investigated the Al–Sc phase diagram over the entire composition and temperature range with thermal and microstructural analysis. Four intermetallic compounds, i.e., Al_3Sc, Al_2Sc, AlSc and $AlSc_2$, were reported. Willey [7], Drits et al. [8], Fujikawa et al. [9], Jo and Fujikawa [10] and Kononenko and Golubev [11] all conducted a study in the Al-rich part of the phase diagram. Enthalpies of formation of the four intermetallic compounds have been measured by Pyagai and Vakhobov [12], Jung et al. [13] and Meschel and Kleppa [14]. Considering the experimental technique they employed, data of Jung et al. [13] and Meshcel

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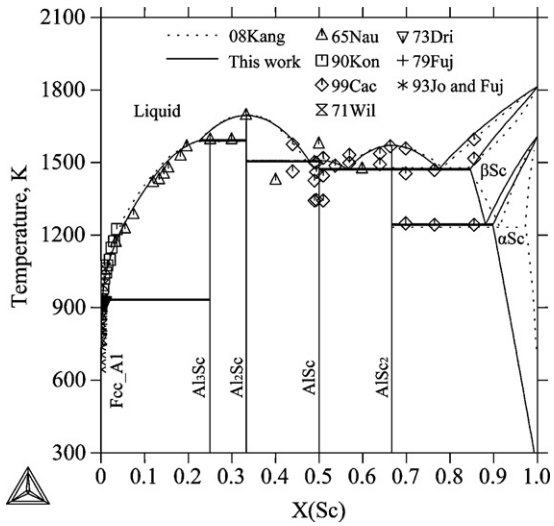

Fig. 1. Calculated phase diagram of the Al–Sc system.

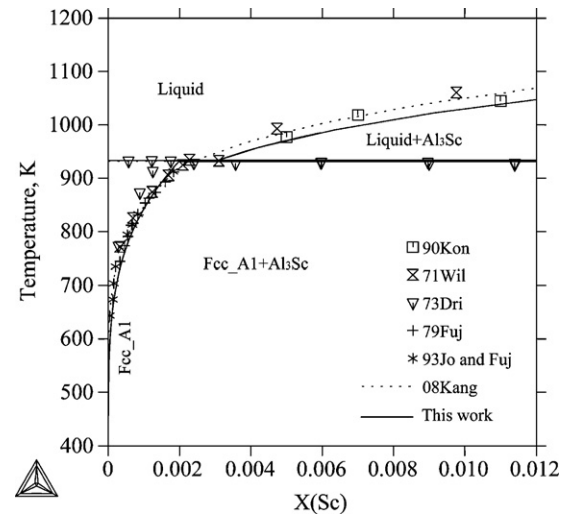

Fig. 2. Calculated enlarged portion of the Al-rich region in the Al–Sc system.

and Kleppa [14] obtained by high temperature calorimetry were thought to be more reliable than that of Pyagai and Vakhobov [12] by acid solution calorimetry. Batalin et al. [15], Zubkov [16] and Pisch et al. [17] measured the enthalpy of mixing of the liquid at 1700 K, 1073 K and 1123 K, respectively. However, their data show a large scatter. The enthalpy data at 1873 K from Litovskii et al. [18] were used in this optimization.

More recently, Cacciamani et al. [19] carried out an examination of the phase equilibria over 40–100 at.% Sc, and the enthalpies of formation of all intermetallic compounds were determined by direct reaction drop calorimetry. Two new invariant reactions were discovered in the Sc-rich part of the diagram: $L \rightarrow \beta Sc + AlSc_2$ at 1185 °C and $\beta Sc \rightarrow AlSc_2 + \alpha Sc$ at 970 °C.

This system has been thermodynamically assessed by Murray [20] and by Cacciamani et al. [19]. In Murray's work [20], the up-to-date lattice stability parameters were not used; while in the paper of Cacciamani et al. [19], the optimized thermodynamic parameters of the solution phases such as liquid, Fcc_Al, Bcc_Sc and Hcp_Sc were not given. Last year, to take account of short-range ordering, Kang et al. [21] remodeled the system and the modified quasi-chemical model was used for the liquid phase. The invariant reaction $\beta Sc \leftrightarrow AlSc_2 + \alpha Sc$ in the Sc-rich part calculated by Kang et al. [21] shows a big difference from that measured by Cacciamani et al. [19] (Fig. 1).

2.2. The Cu–Sc binary system

Phase equilibria of the Cu–Sc system have been investigated by Savitskii et al. [22] and Markiv et al. [23]. Three intermetallic compounds existed, i.e., Cu_4Sc, Cu_2Sc and CuSc. It has been reported that the solubility of Sc in Cu was about 0.5 at.% Sc and that of Cu in Sc was a bit smaller than 1 at.% Cu. The enthalpies of mixing of liquid alloys at 1373 K and 1873 K were studied by Watanabe and Kleppa [24] and Esin et al. [25], respectively. Using high temperature calorimetry, Watanabe and Kleppa [24] also investigated the enthalpies of formation of the three intermetallic compounds at 298 K. With the electromotive force method, Yamshchikov et al. [26] and Goncharuk and Sidorko [27] measured enthalpies of formation of the three compounds between 650 K and 1049 K and at 900 K, respectively. Since calorimetry is a direct method comparing with the EMF method, data of Watanabe and Kleppa [24] were used in this optimization.

Recently, Turchanin [28] has developed a thermodynamic description of the Cu–Sc system, using ideal associated solution model for the liquid phase. With his optimized parameters, the invariant reaction in the Sc-rich side was calculated to be $Liquid + \beta Sc \rightarrow \alpha Sc$. For lack of experimental data, it could be right either for $Liquid + \beta Sc \rightarrow \alpha Sc$ or $Liquid + \alpha Sc \rightarrow \beta Sc$. However, according to most of the Cu-RE phase diagram in the RE-rich region, the latter is more common and is adopted in this work.

One thing needs to be noted is that Savitskii et al. [22] reported the peritectic formation of Cu_2Sc, whereas Markiv et al. [23] reported the congruent formation of it. Since conterminous parts of the liquidus lines of Cu_4Sc and Cu_2Sc are not reasonable as mentioned by Okamoto and Massalski [29], Turchanin [28] treated the formation of Cu_2Sc as peritectic, which is the same in this work.

2.3. The Al–Cu binary system

The Al–Cu system was thermodynamically assessed by Saunders et al. [30]. Witusiewicz et al. [31] reassessed the liquid and γ-$D8_3$ phases on the basis of new experimental thermodynamic data. In this work, the thermodynamic parameters of Witusiewicz et al. [31] were adopted.

2.4. The Al–Cu–Sc ternary system

One ternary compound τ-$Al_{8-x}Cu_{4+x}Sc$ ($0 \leq x \leq 2.6$), with a $ThMn_{12}$ structure, has been reported in the Al-rich corner of the Al–Cu–Sc phase diagram by Kharakterova [32]. He also determined two isothermal sections at 450 °C and 500 °C. Two partial isopleth sections at 20 wt.% Cu with Sc up to 6 wt.%, and from 40 wt.% Cu, 60 wt.% Al to 2 wt.% Sc, 98 wt.% Al, were given by Kharakterova and Kobatkina [33] and Toropova et al. [34], respectively. The liquidus surface in the Al-rich part of the diagram has been established [34], including two invariant reactions $Liquid \rightarrow Fcc_Al + \theta + \tau_1$ at 546 °C and $Liquid + Al_3Sc \rightarrow Fcc_Al + \tau_1$ at 572 °C.

3. Thermodynamic model

3.1. Solution phases

Substitutional solution model was used to describe all the solution phases such as Liquid, Fcc, Bcc and Hcp. The mole Gibbs energy can be expressed as a sum of Gibbs energy of pure elements, Gibbs energy contributed from ideal entropy of mixing and excess Gibbs

energy, i.e.,

$$G^{\phi} = \sum_{i=\mathrm{Al,Cu,Sc}} x_i {}^0G_i^{\phi} + RT \sum_{i=\mathrm{Al,Cu,Sc}} x_i \ln(x_i) + {}^{\mathrm{ex}}G^{\phi} \quad (1)$$

where ϕ denotes the solution phase, x_i denotes the mole fraction of component i(i = Al, Cu, Sc), and ${}^0G_i^{\phi}$ is the molar Gibbs energy of pure element i in the structure of phase ϕ, which is taken from the SGTE data compiled by Dinsdale [35]. The excess term is written as a Redlich-Kister polynomial, i.e.,

$$\begin{aligned} {}^{ex}G^{\phi} = & x_{\mathrm{Al}}x_{\mathrm{Cu}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Cu}})^{j(j)} L^{\phi}_{\mathrm{Al,Cu}} + x_{\mathrm{Al}}x_{\mathrm{Sc}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Sc}})^{j(j)} L^{\phi}_{\mathrm{Al,Sc}} \\ & + x_{\mathrm{Cu}}x_{\mathrm{Sc}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Cu}} - x_{\mathrm{Sc}})^{j(j)} L^{\phi}_{\mathrm{Cu,Sc}} + x_{\mathrm{Al}}x_{\mathrm{Cu}}x_{\mathrm{Sc}} L^{\phi}_{\mathrm{Al,Cu,Sc}} \end{aligned} \quad (2)$$

$${}^{(j)}L^{\phi} \text{ is expressed as } {}^{(j)}L^{\phi} = A_j + B_j T \quad (3)$$

where A_i and B_j are the parameters to be optimized.

$L^{\phi}_{\mathrm{Al,Cu,Sc}}$ is the ternary interaction parameter, which is set to zero for Fcc, Bcc and Hcp phases for lack of experimental data. ${}^0L^{\mathrm{Liq}}_{\mathrm{Al,Cu,Sc}}$ is given a negative number for a better fit to the experimental data of the Al-rich part.

3.2. Binary and ternary intermetallic phases

All binary intermetallic compounds are treated as stoichiometric phases, denoted as M_xN_y (M denotes Al or Cu, N denotes Sc). According to Neumann–Kopp rule, the Gibbs energy of M_xN_y phase is represented as

$$G_{\mathrm{M_xN_y}} = \frac{x}{x+y} {}^0G_{\mathrm{M}}^{\mathrm{Fcc}} + \frac{y}{x+y} {}^0G_{\mathrm{N}}^{\mathrm{Hcp}} + A + BT \quad (4)$$

where A and B are the parameters to be optimized in this work. $A + BT$ is considered as Gibbs energy of formation of M_xN_y.

The ternary compound τ-$Al_{8-x}Cu_{4+x}Sc$ ($0 \le x \le 2.6$) was modeled as (Al, Cu)$_{0.6154}$Cu$_{0.3077}$Sc$_{0.0769}$ (Model 1), since Al_8Cu_4Sc is the most stable compound as suggested by Kang and Chen [36]. The Gibbs energy is expressed as

$$\begin{aligned} G_{(\mathrm{Al,Cu})_{0.6154}\mathrm{Cu}_{0.3077}\mathrm{Sc}_{0.0769}} = & y'_{\mathrm{Al}} G_{\mathrm{Al:Cu:Sc}} + y'_{\mathrm{Cu}} G_{\mathrm{Cu:Cu:Sc}} \\ & + 0.6154RT(y'_{\mathrm{Al}} \ln y'_{\mathrm{Al}} + y'_{\mathrm{Cu}} \ln y'_{\mathrm{Cu}}) \\ & + y'_{\mathrm{Al}} y'_{\mathrm{Cu}} \left(\sum_{j=0,1\ldots} {}^jL_{\mathrm{Al,Cu:Cu:Sc}} (y'_{\mathrm{Al}} - y'_{\mathrm{Cu}})^j \right) \end{aligned} \quad (5)$$

where

$$G_{\mathrm{Al:Cu:Sc}} = 0.6154 {}^0G_{\mathrm{Al}}^{\mathrm{Fcc}} + 0.3077 {}^0G_{\mathrm{Cu}}^{\mathrm{Fcc}} + 0.0769 {}^0G_{\mathrm{Sc}}^{\mathrm{Hcp}} + A_1 + B_1 T \quad (6)$$

$$G_{\mathrm{Cu:Cu:Sc}} = 0.9231 {}^0G_{\mathrm{Cu}}^{\mathrm{Fcc}} + 0.0769 {}^0G_{\mathrm{Sc}}^{\mathrm{Hcp}} + A_2 + B_2 T \quad (7)$$

$L_{\mathrm{Al,Cu:Cu:Sc}}$ is the interaction parameter between Al and Cu in the first sublattice.

In order to give a better description of the crystal structure of τ phase, another model (Al, Cu)$_{0.6154}$(Al, Cu)$_{0.3077}$Sc$_{0.0769}$ (Model 2) was used. The Gibbs energy is expressed as

$$\begin{aligned} G_{(\mathrm{Al,Cu})_{0.6154}(\mathrm{Al,Cu})_{0.3077}\mathrm{Sc}_{0.0769}} = & y'_{\mathrm{Al}} y''_{\mathrm{Al}} G_{\mathrm{Al:Al:Sc}} + y'_{\mathrm{Al}} y''_{\mathrm{Cu}} G_{\mathrm{Al:Cu:Sc}} \\ & + y'_{\mathrm{Cu}} y''_{\mathrm{Al}} G_{\mathrm{Cu:Al:Sc}} \\ & + y'_{\mathrm{Cu}} y''_{\mathrm{Cu}} G_{\mathrm{Cu:Cu:Sc}} + 0.6154RT(y'_{\mathrm{Al}} \ln y'_{\mathrm{Al}} + y'_{\mathrm{Cu}} \ln y'_{\mathrm{Cu}}) \\ & + 0.3077RT(y''_{\mathrm{Al}} \ln y''_{\mathrm{Al}} + y''_{\mathrm{Cu}} \ln y''_{\mathrm{Cu}}) \\ & + y'_{\mathrm{Al}} y'_{\mathrm{Cu}} \left(\sum_{j=0,1\ldots} {}^jL_{\mathrm{Al,Cu:Al:Sc}} (y'_{\mathrm{Al}} - y'_{\mathrm{Cu}})^j \right) \\ & + y'_{\mathrm{Al}} y'_{\mathrm{Cu}} \left(\sum_{j=0,1\ldots} {}^jL_{\mathrm{Al,Cu:Cu:Sc}} (y'_{\mathrm{Al}} - y'_{\mathrm{Cu}})^j \right) \\ & + y''_{\mathrm{Al}} y''_{\mathrm{Cu}} \left(\sum_{j=0,1\ldots} {}^jL_{\mathrm{Al:Al,Cu:Sc}} (y''_{\mathrm{Al}} - y''_{\mathrm{Cu}})^j \right) \\ & + y''_{\mathrm{Al}} y''_{\mathrm{Cu}} \left(\sum_{j=0,1\ldots} {}^jL_{\mathrm{Cu:Al,Cu:Sc}} (y''_{\mathrm{Al}} - y''_{\mathrm{Cu}})^j \right) \end{aligned} \quad (8)$$

The formulations of $G_{\mathrm{Al:Al:Sc}}$, $G_{\mathrm{Al:Cu:Sc}}$, $G_{\mathrm{Cu:Al:Sc}}$ and $G_{\mathrm{Cu:Cu:Sc}}$ are like the Eqs. (6) and (7). $L_{\mathrm{Al,Cu:Al:Sc}}$ and $L_{\mathrm{Al,Cu:Cu:Sc}}$ are the interaction parameters between Al and Cu in the first sublattice, while $L_{\mathrm{Al:Al:Cu:Sc}}$ and $L_{\mathrm{Cu;Al:Cu:Sc}}$ are those in the second sublattice.

4. Optimization results and discussion

The optimization work was performed by Parrot module in the Thermo_calc program developed by Sundman et al. [37]. During the assessment, a certain weight was given for each piece of selected information by personal judgment, and a trial and error method was applied for verification until most of the selected experimental information was reproduced well. All the parameters optimized in this work are listed in Table 1.

4.1. The Al–Sc and Cu–Sc binary system

As shown in Figs. 1 and 2, the calculated Al–Sc phase diagram agrees well with the experimental data [6–11,19]. The calculated invariant reactions along with those by Cacciamani et al. [19] and Kang et al. [21] are summarized in Table 2. The calculated partial enthalpies of mixing of Al and Sc in liquid Al–Sc alloys at 1873 K, enthalpies of mixing of liquid alloys at 1873 K and enthalpies of formation of intermetallic compounds at 298 K are shown in Figs. 3–5, respectively. All the calculated results agree well with the experimental data.

Fig. 6 shows the calculated phase diagram of the Cu–Sc system compared with the experimental data and the previous work [28]. All the calculated temperatures agree well with the experimental data within the range of 4 K. The calculated invariant reactions along with those of Turchanin [28] are shown in Table 3. In Figs. 7 and 8, enthalpies of mixing of liquid Cu–Sc alloys at 1373 K and 1873 K calculated from the present work were compared with the previous calculation from Turchanin [28] and the experimental data [24,25,28]. As to the enthalpy of formation of intermetallic compounds, the calculated results exhibit a good agreement with the calorimetry data of Watanabe and Kleppa [24] while deviating from the EMF data [26,27], as shown in Fig. 9. This result is acceptable as discussed in Section 2.2.

Table 1
Thermodynamic parameters of the Al–Cu–Sc system.

Phase	Thermodynamic parameters	Ref.
Liquid Model: (Al, Cu, Sc)	${}^{0}L^{Liq}_{Al,Cu} = -67094 + 8.555T$	[31]
	${}^{1}L^{Liq}_{Al,Cu} = 32148 - 7.118T$	[31]
	${}^{2}L^{Liq}_{Al,Cu} = 5915 - 5.889T$	[31]
	${}^{3}L^{Liq}_{Al,Cu} = -8175 + 6.049T$	[31]
	${}^{0}L^{Liq}_{Al,Sc} = -131778 + 3.639T$	This work
	${}^{1}L^{Liq}_{Al,Sc} = -20126 + 8.877T$	This work
	${}^{2}L^{Liq}_{Al,Sc} = 50985 - 24.917T$	This work
	${}^{0}L^{Liq}_{Cu,Sc} = -96000 + 45.799T$	This work
	${}^{1}L^{Liq}_{Cu,Sc} = -12417 + 5.869T$	This work
	${}^{2}L^{Liq}_{Cu,Sc} = 12271 - 9.715T$	This work
	${}^{0}L^{Liq}_{Al,Cu,Sc} = -58000$	This work
Fcc Model: (Al, Cu, Sc)	${}^{0}L^{Fcc}_{Al,Cu} = -53520 + 2T$	[31]
	${}^{1}L^{Fcc}_{Al,Cu} = 38590 - 2T$	[31]
	${}^{2}L^{Fcc}_{Al,Cu} = 1170$	[31]
	${}^{0}L^{Fcc}_{Al,Sc} = -109500 + 2.5T$	This work
	${}^{0}L^{Fcc}_{Cu,Sc} = -3890$	This work
Bcc Model: (Al, Cu, Sc)	${}^{0}L^{Bcc}_{Al,Cu} = -73554 + 4T$	[31]
	${}^{1}L^{Bcc}_{Al,Cu} = 51500 - 11.84T$	[31]
	${}^{0}L^{Bcc}_{Al,Sc} = -113589$	This work
	${}^{0}L^{Bcc}_{Cu,Sc} = 6001$	This work
Hcp Model: (Al, Cu, Sc)	${}^{0}L^{Hcp}_{Al,Sc} = -106397$	This work
	${}^{0}L^{Hcp}_{Cu,Sc} = 14500 - 3T$	This work
θ Model: $(Al)_2(Al, Cu)_1$	$G^{\theta}_{Al:Al} = 30249 - 14.439T + 3GHSERAL$	[31]
	$G^{\theta}_{Al:Cu} = -47406 + 6.75T + 2GHSERAL + GHSERCU$	[31]
	${}^{0}L^{\theta}_{Al:Al,Cu} = 2211$	[31]
η Model: $(Al, Cu)_1(Cu)_1$	$G^{\eta}_{Al:Cu} = -40560 + 3.14T + GHSERAL + GHSERCU$	[31]
	$G^{\eta}_{Cu:Cu} = +8034 - 2.51T + 2GHSERCU$	[31]
	${}^{0}L^{\eta}_{Al,Cu:Cu} = -25740 - 20T$	[31]
ε Model: $(Al, Cu)_1(Cu)_1$	$G^{\varepsilon}_{Al:Cu} = -36976 + 1.2T + GHSERAL + GHSERCU$	[31]
	$G^{\varepsilon}_{Cu:Cu} = +8034 - 2.51T + 2GHSERCU$	[31]
	${}^{0}L^{\varepsilon}_{Al,Cu:Cu} = +7600 - 24T$	[31]
	${}^{1}L^{\varepsilon}_{Al,Cu:Cu} = -72000$	[31]
ζ Model: $(Al)_9(Cu)_{11}$	$G^{\zeta}_{Al:Cu} = -420000 + 18T + 9GHSERAL + 11GHSERCU$	[31]
δ Model: $(Al)_2(Cu)_3$	$G^{\delta}_{Al:Cu} = -106700 + 3T + 2GHSERAL + 3GHSERCU$	[31]
γ_D8$_3$ Model: $(Al)_4(Al, Cu)_1(Cu)_8$	$G^{\gamma_D8_3}_{Al:Al:Cu} = -277739 + 215T - 30T\ln(T) + 5GHSERAL + 8GHSERCU$	[31]
	$G^{\gamma_D8_3}_{Al:Cu:Cu} = -280501 + 379.6T - 52T\ln(T) + 4GHSERAL + 9GHSERCU$	[31]
γ Model: $(Al)_4(Al, Cu)_1(Cu)_8$	$G^{\gamma}_{Al:Al:Cu} = -219258 - 45.5T + 5GHSERAL + 8GHSERCU$	[31]
	$G^{\gamma}_{Al:Al:Cu} = -200460 - 58.5T + 4GHSERAL + 9GHSERCU$	[31]
Al_3Sc Model: $(Al)_3(Sc)_1$	$G^{Al_3Sc}_{Al:Sc} = -156700 + 7.2T + 3GHSERAL + GHSERSC$	This work
Al_2Sc Model: $(Al)_2(Sc)_1$	$G^{Al_2Sc}_{Al:Sc} = -142500 + 8.8T + 2GHSERAL + GHSERSC$	This work
AlSc Model: $(Al)_1(Sc)_1$	$G^{AlSc}_{Al:Sc} = -94100 + 6.4T + GHSERAL + GHSERSC$	This work
$AlSc_2$ Model: $(Al)_1(Sc)_2$	$G^{AlSc_2}_{Al:Sc} = -112484 + 1.6T + GHSERAL + 2GHSERSC$	This work
Cu_4Sc Model: $(Cu)_4(Sc)_1$	$G^{Cu_4Sc}_{Cu:Sc} = -64156 + 13T + 4GHSERCU + GHSERSC$	This work
Cu_2Sc Model: $(Cu)_2(Sc)_1$	$G^{Cu_2Sc}_{Cu:Sc} = -58500 + 17T + 2GHSERCU + GHSERSC$	This work
CuSc Model: $(Cu)_1(Sc)_1$	$G^{CuSc}_{Cu:Sc} = -44705 + 13.5T + GHSERCU + GHSERSC$	This work
τ Model 1: $(Al, Cu)_{0.6154}(Cu)_{0.3077}(Sc)_{0.0769}$	$G^{\tau}_{Al:Cu:Sc} = -25280 + 0.6154GHSERAL + 0.3077GHSERCU + 0.0769GHSERSC$	This work
	$G^{\tau}_{Cu:Cu:Sc} = 2500 + 0.9231GHSERCU + 0.0769GHSERSC$	This work
	$G^{\tau}_{Al,Cu:Cu:Sc} = -32900$	This work
τ Model 2: $(Al, Cu)_{0.6154}(Al, Cu)_{0.3077}(Sc)_{0.0769}$	$G^{\tau}_{Al:Cu:Sc} = -25220 + 0.6154GHSERAL + 0.3077GHSERCU + 0.0769GHSERSC$	This work
	$G^{\tau}_{Cu:Al:Sc} = -2800 + 0.6154GHSERCU + 0.3077GHSERAL + 0.0769GHSERSC$	This work
	$G^{\tau}_{Al:Al:Sc} = -2800 + 0.9231GHSERAL + 0.0769GHSERSC$	This work
	$G^{\tau}_{Cu:Cu:Sc} = 2500 + 0.9231GHSERCU + 0.0769GHSERSC$	This work
	$G^{\tau}_{Al,Cu:Al:Sc} = G^{\tau}_{Al,Cu:Cu:Sc} = -33160$	This work
	$G^{\tau}_{Al:Al,Cu:Sc} = G^{\tau}_{Cu:Al,Cu:Sc} = -2800$	This work

4.2. The Al–Cu–Sc ternary system in the Al-rich corner

The calculated partial isopleth sections at 20 wt.% Cu and from 40 wt.% Cu, 60 wt.% Al to 2 wt.% Sc, 98 wt.% Al, as shown in Figs. 10 and 11, respectively, are in good agreement with those reported by Kharakterova and Kobatkina [33] and Toropova et al. [34]. Comparing with Model 1, the calculated temperature of the invariant reaction Liquid + Al_3Sc ↔ τ + Fcc_Al increases 2 °C by Model 2 and the phase field of Fcc_Al + τ shifts a little to the right. Except these, the phase boundaries calculated with the two models

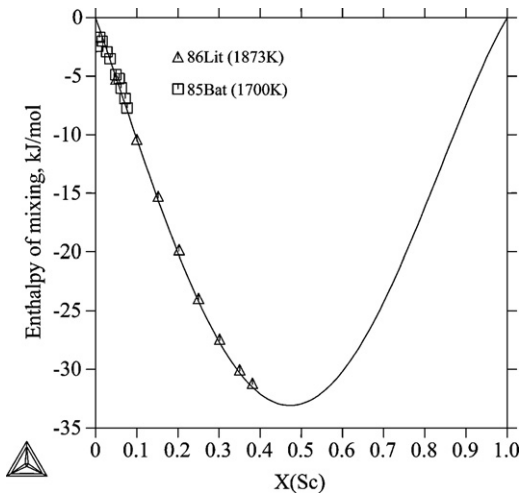

Fig. 3. Calculated enthalpies of mixing of the liquid Al–Sc alloys at 1873 K.

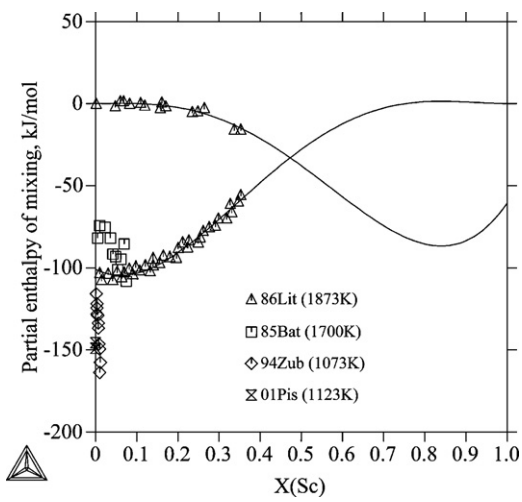

Fig. 4. Calculated partial enthalpies of mixing of Al and Sc in liquid Al–Sc alloys at 1873 K.

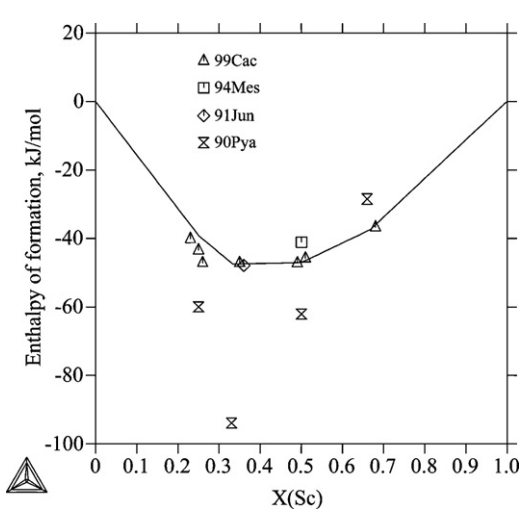

Fig. 5. Calculated enthalpies of formation of the Al–Sc alloys at 298 K.

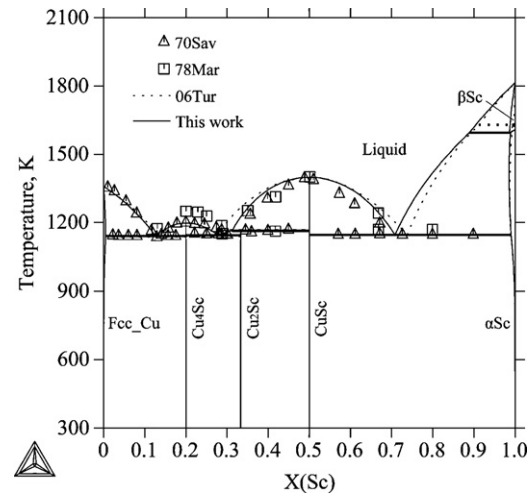

Fig. 6. Calculated phase diagram of the Cu–Sc system.

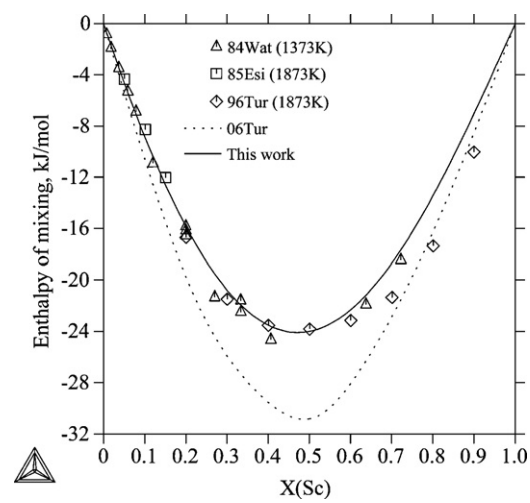

Fig. 7. Calculated enthalpies of mixing of the liquid Cu–Sc alloys at 1373 K.

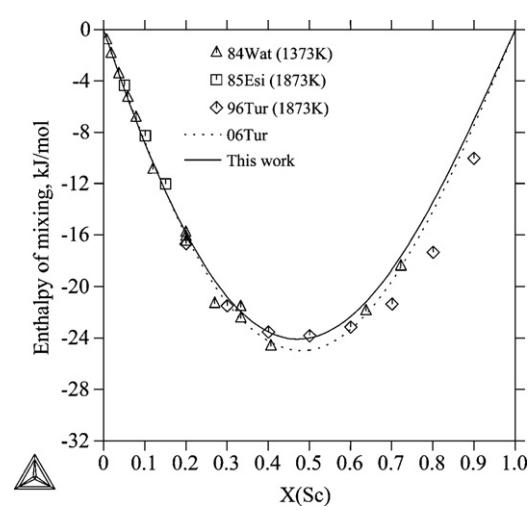

Fig. 8. Calculated enthalpies of mixing of the liquid Cu–Sc alloys at 1873 K.

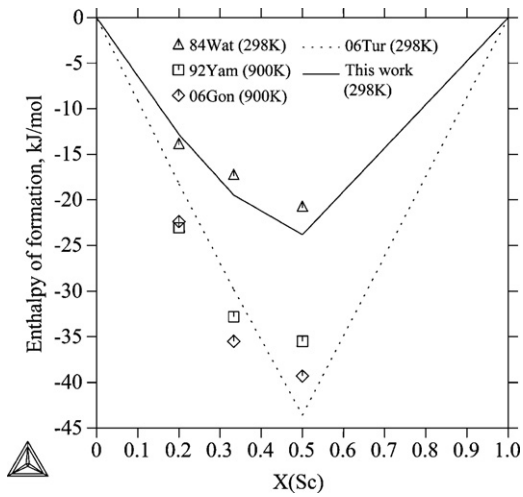

Fig. 9. Calculated enthalpies of formation of the Cu–Sc alloys at 298 K.

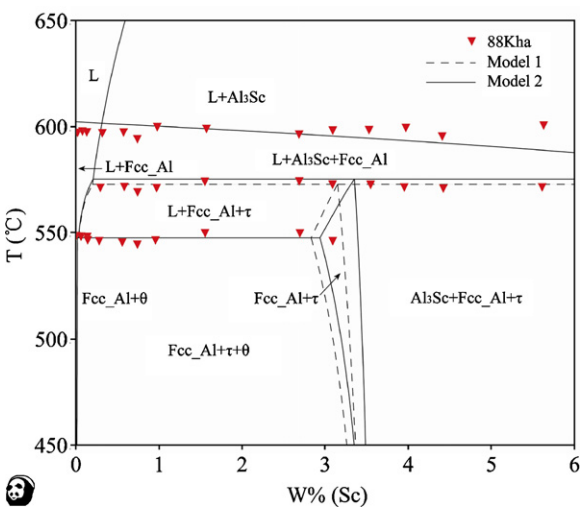

Fig. 10. Calculated isoplethic section at 20 wt.% Cu.

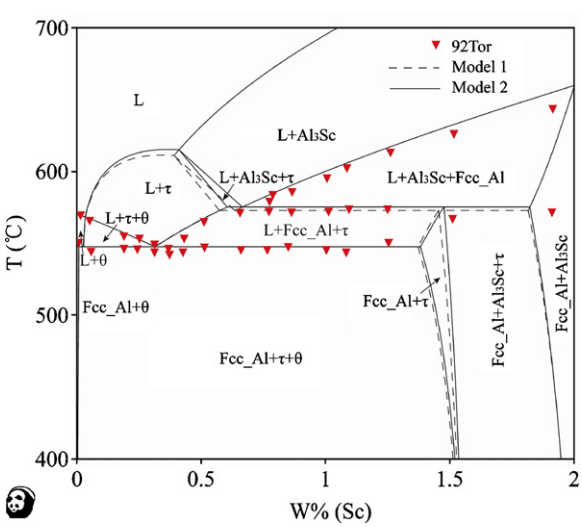

Fig. 11. Calculated isoplethic section of the Al-rich corner from 40 wt.% Cu, 60 wt.% Al to 2 wt.% Sc, 98 wt.% Al.

Table 2
Invariant reactions in the Al–Sc system.

Reaction	T (K)	Phase composition X_{Sc}			Ref.
Liquid ↔ Fcc_Al + Al_3Sc	932	0.00313	0.00228	0.250	[7]
	929.5	0.00566	0.00415	0.250	[19]
	933				[21]
	933	0.00303	0.00200	0.250	This work
Liquid + Al_2Sc ↔ Al_3Sc	1593	0.240	0.333	0.250	[6]
	1556.0	0.207	0.333	0.250	[19]
	1589				[21]
	1591	0.226	0.333	0.250	This work
Liquid ↔ Al_2Sc	1693	0.333	0.333		[6]
	1664.5	0.333	0.333		[19]
	1698				[21]
	1694	0.333	0.333		This work
Liquid ↔ Al_2Sc + AlSc	1463	0.450	0.333	0.500	[19]
	1506.6	0.491	0.333	0.500	[19]
	1508				[21]
	1505	0.478	0.333	0.500	This work
Liquid ↔ AlSc	1513	0.500	0.500		[19]
	1507.1	0.500	0.500		[19]
	1519				[21]
	1509	0.500	0.500		This work
Liquid ↔ AlSc + $AlSc_2$	1468	>0.670	0.500	0.667	[6]
	1478.1	0.566	0.500	0.667	[19]
	1474				[21]
	1472	0.565	0.500	0.667	This work
Liquid ↔ $AlSc_2$	1573	0.667	0.667		[19]
	1560.0	0.667	0.667		[19]
	1574				[21]
	1571	0.667	0.667		This work
Liquid ↔ $AlSc_2$ + βSc	1458	0.770	0.667	0.810	[19]
	1460.6	0.775	0.667	0.798	[19]
	1479				[21]
	1472	0.774	0.667	0.846	This work
βSc ↔ $AlSc_2$ + αSc	1243	0.850	0.667	0.930	[19]
	1250.1	0.872	0.667	0.908	[19]
	1233				[21]
	1242	0.881	0.667	0.898	This work

Table 3
Invariant reactions in the Cu–Sc system.

Reaction	T (K)	Phase composition X_{Sc}			Ref.
Liquid ↔ Fcc_Cu + Cu_4Sc	1138	0.130	0.00500	0.200	[22]
	1140	0.132	0.00480	0.200	[28]
	1141	0.126	0.00500	0.200	This work
Liquid ↔ Cu_4Sc	1198	0.200	0.200		[22]
	1198	0.200	0.200		[28]
	1198	0.200	0.200		This work
Liquid ↔ Cu_4Sc + Cu_2Sc	1148	0.300	0.200	0.333	[22]
	1141	0.263	0.200	0.333	[28]
	1144	0.278	0.200	0.333	This work
Liquid + CuSc ↔ Cu_2Sc	1163	–	0.500	0.333	[22]
	1165	0.285	0.500	0.333	[28]
	1163	0.312	0.500	0.333	This work
Liquid ↔ CuSc	1398	0.500	0.500		[22]
	1399	0.500	0.500		[28]
	1399	0.500	0.500		This work
Liquid ↔ CuSc + αSc	1148	0.710	0.500	0.990	[22]
	1145	0.738	0.500	0.989	[28]
	1145	0.708	0.500	0.990	This work
Liquid + βSc ↔ αSc	1641	0.919	0.994	1.000	[28]
Liquid + αSc ↔ βSc	1595	0.889	0.989	0.986	This work

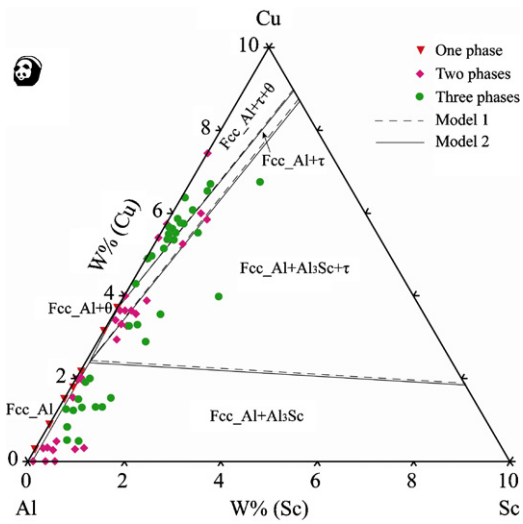

Fig. 12. Calculated isothermal section of the Al-rich corner at 500 °C along with the experimental data from Kharakterova [32].

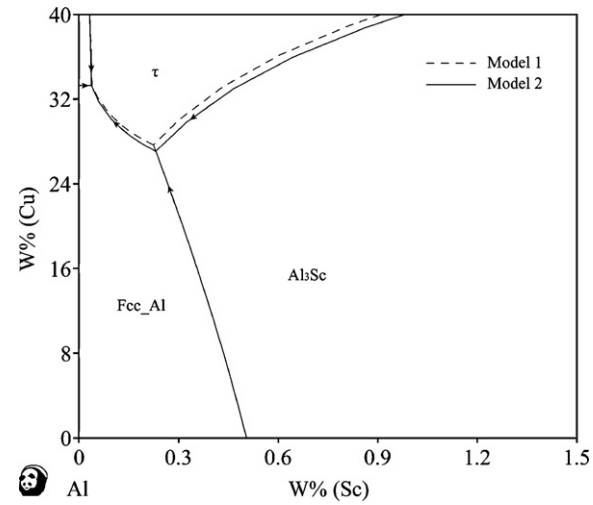

Fig. 13. Calculated liquidus surface of the Al-rich corner.

are almost in the same place. The calculated isothermal section at 500 °C of the Al-rich region is shown in Fig. 12. According to Kharakterova [32], the maximum solubility for Al in the Fcc_Al + τ + Al_2Cu three-phase equilibrium is 3.75 wt.% Cu, 0.025 wt.% Sc at 500 °C, and it was calculated to be 3.77 wt.% Cu, 0.020 wt.% Sc using Model 1. The maximum solubility for Al in the Fcc_Al + τ + Al_3Sc three-phase equilibrium is 0.56 wt.% Cu, 0.07 wt.% Sc at 500 °C according to the report [32], while it was calculated to be 2.43 wt.% Cu, 0.077 wt.% Sc. Since the deviation of the composition of Cu is a bit larger, Model 2 was used to analyze whether this was due to the impropriety of the model. The calculated composition by Model 2 was 2.38 wt.% Cu, only a little better than that by Model 1. The discrepancy was attributed to the inconsistency of the solubility in the ternary system and that in the Al–Cu binary system. According to the optimized Al–Cu phase diagram by Witusiewicz et al. [31], the solubility of Cu in Fcc_Al is about 3.8 wt.% Cu at 500 °C. To abrupt change of the solubility of Cu in the ternary and binary system is not reasonable. Concerning Kharakterova's experiment [32], he annealed the Al–Cu–Sc alloys for 500 h when measured the isothermal section at 500 °C. It is possible that the samples did not reach the equilibrium states. Further experiments are needed for accurate experimental measurement. Partial liquidus surface of the Al-rich corner was calculated, as shown in Fig. 13. The calculated compositions of invariant reactions by Model 2 are closer to the experimental data. Table 4 lists the comparison of calculated results by the two models and the experimental data. It can be seen that the difference is small.

Table 4
Calculated invariant reactions in the Al-rich part compared with the experimental data.

Type	Reaction	Concentrations in liquid		T (°C)	Ref.
		Cu (wt.%)	Sc (wt.%)		
E	Liquid ↔ τ + Fcc_Al + θ	31.2	0.07	546	[34]
		33.3	0.04	547	This work (Model 1)
		33.3	0.04	547	This work (Model 2)
U	Liquid + Al_3Sc ↔ τ + Fcc_Al	25.0	0.22	572	[34]
		27.6	0.22	573	This work (Model 1)
		27.0	0.23	575	This work (Model 2)

5. Conclusions

Using the CALPHAD technique, the Al–Sc and Cu–Sc binary systems were reassessed and the calculated results showed good agreement with the experimental data. Incorporating the thermodynamic parameters of the three binary sub-systems, the Al–Cu–Sc system in the Al-rich corner was thermodynamically assessed. There are two invariant reactions, Liquid ↔ τ + Fcc_Al + θ and Liquid + Al_3Sc ↔ τ + Fcc_Al, calculated in this work. Two models, $(Al, Cu)_{0.6154}Cu_{0.3077}Sc_{0.0769}$ and $(Al, Cu)_{0.6154}(Al, Cu)_{0.3077}Sc_{0.0769}$, were used for the ternary compound $Al_{8-x}Cu_{4+x}Sc$. The calculated liquidus projection and isothermal and isopleth sections with the two models showed little difference and both agreed well with most of the experimental data. However, there was a descrepancy between the calculation and the experimental data about the maximum solubility for Al in the Fcc_Al + τ + Al_3Sc three-phase equilibrium. The result calculated using Model 1 was 2.43 wt.% Cu, whereas the reported experimental data is 0.56 wt.% Cu. When using Model 2, it was calculated as 2.38 wt.% Cu. Since Model 2 describes the crystal structure of $Al_{8-x}Cu_{4+x}$ Sc more specifically, it should be better than Model 1 and this is confirmed by the calculated result mentioned above, although the advantage is slight.

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